

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 15D

CARBOXYLIC ACID DERIVATIVES (ACYL CHLORIDES, ESTERS & POLYESTERS)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: CARBOXYLIC ACID DERIVATIVES AND POLYESTERS
SUB-TOPIC:	15D: Carboxylic Acid Derivatives (Acyl Chlorides, Esters & Polyesters)
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	203 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. Acyl Chloride Reactivity with Water****[4 Marks]****WCH14/01/Jan23/Q21**

Ethanoyl chloride (CH_3COCl) reacts vigorously with water at room temperature.

- (a) Write the balanced equation for the reaction of ethanoyl chloride with water. [1]

- (b) State the observation and name the gaseous product formed. [2]

- (c) Explain why ethanoyl chloride is much more reactive towards water than chlorobenzene. [1]

Q2. 2. Acid vs Alkaline Hydrolysis of Esters**[5 Marks]****WCH14/01/Oct22/Q22**

Compare the acid-catalysed hydrolysis and alkaline hydrolysis (saponification) of methyl propanoate.

- (a) Write the equation for the acid-catalysed hydrolysis of methyl propanoate using dilute H_2SO_4 . [2]

- (b) Write the equation for the alkaline hydrolysis of methyl propanoate using aqueous NaOH . [2]

- (c) Explain why alkaline hydrolysis goes to completion whereas acid hydrolysis is an equilibrium. [1]

Q3. 3. Esterification & Acyl Chloride Reaction Step 3**[3 Marks]****WCH14/01/sample/Q3**

Consider ester derivative 3: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

- (a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

- (b) State one disadvantage of using an acyl chloride. [1]

Q4. 4. Esterification & Acyl Chloride Reaction Step 4**[3 Marks]****WCH14/01/sample/Q4**

Consider ester derivative 4: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

- (a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q5. 5. Esterification & Acyl Chloride Reaction Step 5

[3 Marks]

WCH14/01/Sample/Q5

Consider ester derivative 5: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q6. 6. Esterification & Acyl Chloride Reaction Step 6

[3 Marks]

WCH14/01/Sample/Q6

Consider ester derivative 6: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q7. 7. Esterification & Acyl Chloride Reaction Step 7

[3 Marks]

WCH14/01/Sample/Q7

Consider ester derivative 7: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q8. 8. Esterification & Acyl Chloride Reaction Step 8

[3 Marks]

WCH14/01/Sample/Q8

Consider ester derivative 8: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q9. 9. Esterification & Acyl Chloride Reaction Step 9

[3 Marks]

WCH14/01/sample/Q9

Consider ester derivative 9: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q10. 10. Esterification & Acyl Chloride Reaction Step 10

[3 Marks]

WCH14/01/sample/Q10

Consider ester derivative 10: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q11. 11. Esterification & Acyl Chloride Reaction Step 11

[3 Marks]

WCH14/01/sample/Q11

Consider ester derivative 11: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q12. 12. Esterification & Acyl Chloride Reaction Step 12

[3 Marks]

WCH14/01/sample/Q12

Consider ester derivative 12: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q13. 13. Esterification & Acyl Chloride Reaction Step 13**[3 Marks]****WCH14/01/Sample/Q13**

Consider ester derivative 13: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q14. 14. Esterification & Acyl Chloride Reaction Step 14**[3 Marks]****WCH14/01/Sample/Q14**

Consider ester derivative 14: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q15. 15. Esterification & Acyl Chloride Reaction Step 15**[3 Marks]****WCH14/01/Sample/Q15**

Consider ester derivative 15: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q16. 16. Esterification & Acyl Chloride Reaction Step 16**[3 Marks]****WCH14/01/Sample/Q16**

Consider ester derivative 16: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q17. 17. Esterification & Acyl Chloride Reaction Step 17**[3 Marks]****WCH14/01/Sample/Q17**

Consider ester derivative 17: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q18. 18. Esterification & Acyl Chloride Reaction Step 18**[3 Marks]****WCH14/01/Sample/Q18**

Consider ester derivative 18: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q19. 19. Esterification & Acyl Chloride Reaction Step 19**[3 Marks]****WCH14/01/Sample/Q19**

Consider ester derivative 19: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q20. 20. Esterification & Acyl Chloride Reaction Step 20**[3 Marks]****WCH14/01/Sample/Q20**

Consider ester derivative 20: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q21. 21. Esterification & Acyl Chloride Reaction Step 21**[3 Marks]****WCH14/01/Sample/Q21**

Consider ester derivative 21: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q22. 22. Esterification & Acyl Chloride Reaction Step 22

[3 Marks]

WCH14/01/sample/Q22

Consider ester derivative 22: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q23. 23. Esterification & Acyl Chloride Reaction Step 23

[3 Marks]

WCH14/01/sample/Q23

Consider ester derivative 23: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q24. 24. Esterification & Acyl Chloride Reaction Step 24

[3 Marks]

WCH14/01/sample/Q24

Consider ester derivative 24: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

Q25. 25. Esterification & Acyl Chloride Reaction Step 25

[3 Marks]

WCH14/01/sample/Q25

Consider ester derivative 25: $\text{CH}_3\text{COCl} + \text{R-OH} \rightarrow \text{CH}_3\text{COOR} + \text{HCl}$.

(a) State one advantage of using an acyl chloride rather than a carboxylic acid to prepare an ester. [2]

(b) State one disadvantage of using an acyl chloride. [1]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. A* Challenge: Condensation Polymers & Biodegradability 26****[5 Marks]**
WCH14/01/Hard/Q26

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

- (a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q27. 27. A* Challenge: Condensation Polymers & Biodegradability 27**[5 Marks]**
WCH14/01/Hard/Q27

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

- (a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q28. 28. A* Challenge: Condensation Polymers & Biodegradability 28**[5 Marks]**
WCH14/01/Hard/Q28

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

- (a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q29. 29. A* Challenge: Condensation Polymers & Biodegradability 29**[5 Marks]**
WCH14/01/Hard/Q29

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

- (a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q30. 30. A* Challenge: Condensation Polymers & Biodegradability 30**[5 Marks]**
WCH14/01/Hard/Q30

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

- (a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q31. 31. A* Challenge: Condensation Polymers & Biodegradability 31**[5 Marks]**
WCH14/01/Hard/Q31

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

- (a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q32. 32. A* Challenge: Condensation Polymers & Biodegradability 32**[5 Marks]**
WCH14/01/Hard/Q32

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

(a) Draw the repeat unit of Terylene showing all ester linkages. [2]

(b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q33. 33. A* Challenge: Condensation Polymers & Biodegradability 33**[5 Marks]**
WCH14/01/Hard/Q33

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

(a) Draw the repeat unit of Terylene showing all ester linkages. [2]

(b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q34. 34. A* Challenge: Condensation Polymers & Biodegradability 34**[5 Marks]**
WCH14/01/Hard/Q34

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

(a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q35. 35. A* Challenge: Condensation Polymers & Biodegradability 35

[5 Marks]
WCH14/01/Hard/Q35

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

- (a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q36. 36. A* Challenge: Condensation Polymers & Biodegradability 36

[5 Marks]
WCH14/01/Hard/Q36

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

- (a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q37. 37. A* Challenge: Condensation Polymers & Biodegradability 37

[5 Marks]
WCH14/01/Hard/Q37

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

- (a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q38. 38. A* Challenge: Condensation Polymers & Biodegradability 38

[5 Marks]
WCH14/01/Hard/Q38

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

- (a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q39. 39. A* Challenge: Condensation Polymers & Biodegradability 39

[5 Marks]
WCH14/01/Hard/Q39

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

- (a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q40. 40. A* Challenge: Condensation Polymers & Biodegradability 40

[5 Marks]
WCH14/01/Hard/Q40

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

(a) Draw the repeat unit of Terylene showing all ester linkages. [2]

(b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q41. 41. A* Challenge: Condensation Polymers & Biodegradability 41

[5 Marks]
WCH14/01/Hard/Q41

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

(a) Draw the repeat unit of Terylene showing all ester linkages. [2]

(b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q42. 42. A* Challenge: Condensation Polymers & Biodegradability 42

[5 Marks]
WCH14/01/Hard/Q42

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

(a) Draw the repeat unit of Terylene showing all ester linkages. [2]

(b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q43. 43. A* Challenge: Condensation Polymers & Biodegradability 43

[5 Marks]
WCH14/01/Hard/Q43

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

(a) Draw the repeat unit of Terylene showing all ester linkages. [2]

(b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q44. 44. A* Challenge: Condensation Polymers & Biodegradability 44

[5 Marks]
WCH14/01/Hard/Q44

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

(a) Draw the repeat unit of Terylene showing all ester linkages. [2]

(b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q45. 45. A* Challenge: Condensation Polymers & Biodegradability 45

[5 Marks]
WCH14/01/Hard/Q45

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

(a) Draw the repeat unit of Terylene showing all ester linkages. [2]

(b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q46. 46. A* Challenge: Condensation Polymers & Biodegradability 46**[5 Marks]**
WCH14/01/Hard/Q46

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

- (a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q47. 47. A* Challenge: Condensation Polymers & Biodegradability 47**[5 Marks]**
WCH14/01/Hard/Q47

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

- (a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q48. 48. A* Challenge: Condensation Polymers & Biodegradability 48**[5 Marks]**
WCH14/01/Hard/Q48

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

- (a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q49. 49. A* Challenge: Condensation Polymers & Biodegradability 49**[5 Marks]**
WCH14/01/Hard/Q49

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

- (a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

Q50. 50. A* Challenge: Condensation Polymers & Biodegradability 50**[5 Marks]**
WCH14/01/Hard/Q50

Terylene (PET) is formed by the condensation polymerisation of benzene-1,4-dicarboxylic acid and ethane-1,2-diol.

- (a) Draw the repeat unit of Terylene showing all ester linkages. [2]

- (b) Explain why polyesters are biodegradable in the environment whereas poly(ethene) is non-biodegradable. [3]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 15D

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Acyl Chloride Reactivity with Water 1. (a) $\text{CH}_3\text{COCl} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH} + \text{HCl}$ (1). 1. (b) Misty / steamy white fumes (1); hydrogen chloride gas (HCl) (1). 1. (c) Carbonyl C in CH_3COCl is highly electron deficient (δ^+) attached to two electronegative atoms (O and Cl); chlorobenzene has p-orbital overlap with benzene ring strengthening C-Cl bond (1).	[4]
Q2	Q2: 2. Acid vs Alkaline Hydrolysis of Esters 2. (a) $\text{CH}_3\text{CH}_2\text{COOCH}_3 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{COOH} + \text{CH}_3\text{OH}$ (2). 2. (b) $\text{CH}_3\text{CH}_2\text{COOCH}_3 + \text{NaOH} \rightarrow \text{CH}_3\text{CH}_2\text{COONa} + \text{CH}_3\text{OH}$ (2). 2. (c) Alkaline hydrolysis forms carboxylate ion $\text{CH}_3\text{CH}_2\text{COO}^-$ which is non-reactive towards nucleophilic attack by CH_3OH (1).	[5]
Q3	Q3: 3. Esterification & Acyl Chloride Reaction Step 3 3. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 3. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q4	Q4: 4. Esterification & Acyl Chloride Reaction Step 4 4. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 4. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q5	Q5: 5. Esterification & Acyl Chloride Reaction Step 5 5. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 5. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q6	Q6: 6. Esterification & Acyl Chloride Reaction Step 6 6. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 6. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q7	Q7: 7. Esterification & Acyl Chloride Reaction Step 7 7. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 7. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q8	Q8: 8. Esterification & Acyl Chloride Reaction Step 8 8. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 8. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q9	Q9: 9. Esterification & Acyl Chloride Reaction Step 9 9. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 9. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q10	Q10: 10. Esterification & Acyl Chloride Reaction Step 10 10. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 10. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q11	Q11: 11. Esterification & Acyl Chloride Reaction Step 11 11. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 11. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]

Q12	Q12: 12. Esterification & Acyl Chloride Reaction Step 12 12. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 12. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q13	Q13: 13. Esterification & Acyl Chloride Reaction Step 13 13. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 13. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q14	Q14: 14. Esterification & Acyl Chloride Reaction Step 14 14. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 14. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q15	Q15: 15. Esterification & Acyl Chloride Reaction Step 15 15. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 15. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q16	Q16: 16. Esterification & Acyl Chloride Reaction Step 16 16. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 16. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q17	Q17: 17. Esterification & Acyl Chloride Reaction Step 17 17. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 17. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q18	Q18: 18. Esterification & Acyl Chloride Reaction Step 18 18. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 18. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q19	Q19: 19. Esterification & Acyl Chloride Reaction Step 19 19. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 19. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q20	Q20: 20. Esterification & Acyl Chloride Reaction Step 20 20. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 20. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q21	Q21: 21. Esterification & Acyl Chloride Reaction Step 21 21. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 21. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q22	Q22: 22. Esterification & Acyl Chloride Reaction Step 22 22. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 22. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q23	Q23: 23. Esterification & Acyl Chloride Reaction Step 23 23. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 23. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q24	Q24: 24. Esterification & Acyl Chloride Reaction Step 24 24. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 24. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]

Q25	Q25: 25. Esterification & Acyl Chloride Reaction Step 25 25. (a) Reaction is fast at room temperature (1) and goes to 100% completion (irreversible) (1). 25. (b) Produces toxic / corrosive HCl gas fumes (1).	[3]
Q26	Q26: 26. A* Challenge: Condensation Polymers & Biodegradability 26 26. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 26. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q27	Q27: 27. A* Challenge: Condensation Polymers & Biodegradability 27 27. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 27. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q28	Q28: 28. A* Challenge: Condensation Polymers & Biodegradability 28 28. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 28. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q29	Q29: 29. A* Challenge: Condensation Polymers & Biodegradability 29 29. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 29. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q30	Q30: 30. A* Challenge: Condensation Polymers & Biodegradability 30 30. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 30. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q31	Q31: 31. A* Challenge: Condensation Polymers & Biodegradability 31 31. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 31. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q32	Q32: 32. A* Challenge: Condensation Polymers & Biodegradability 32 32. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 32. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q33	Q33: 33. A* Challenge: Condensation Polymers & Biodegradability 33 33. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 33. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q34	Q34: 34. A* Challenge: Condensation Polymers & Biodegradability 34 34. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 34. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q35	Q35: 35. A* Challenge: Condensation Polymers & Biodegradability 35 35. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 35. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]

Q36	Q36: 36. A* Challenge: Condensation Polymers & Biodegradability 36 36. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 36. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q37	Q37: 37. A* Challenge: Condensation Polymers & Biodegradability 37 37. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 37. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q38	Q38: 38. A* Challenge: Condensation Polymers & Biodegradability 38 38. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 38. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q39	Q39: 39. A* Challenge: Condensation Polymers & Biodegradability 39 39. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 39. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q40	Q40: 40. A* Challenge: Condensation Polymers & Biodegradability 40 40. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 40. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q41	Q41: 41. A* Challenge: Condensation Polymers & Biodegradability 41 41. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 41. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q42	Q42: 42. A* Challenge: Condensation Polymers & Biodegradability 42 42. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 42. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q43	Q43: 43. A* Challenge: Condensation Polymers & Biodegradability 43 43. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 43. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q44	Q44: 44. A* Challenge: Condensation Polymers & Biodegradability 44 44. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 44. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q45	Q45: 45. A* Challenge: Condensation Polymers & Biodegradability 45 45. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 45. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]
Q46	Q46: 46. A* Challenge: Condensation Polymers & Biodegradability 46 46. (a) $-O-CO-C_6H_4-CO-O-CH_2-CH_2-$ with open continuation bonds at ends (2). 46. (b) Polyesters contain polar $C=O$ and $C-O$ ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar $C-C$ and $C-H$ bonds resistant to chemical and enzymatic attack (1).	[5]

Q47	Q47: 47. A* Challenge: Condensation Polymers & Biodegradability 47 47. (a) $\text{-O-CO-C}_6\text{H}_4\text{-CO-O-CH}_2\text{-CH}_2\text{-}$ with open continuation bonds at ends (2). 47. (b) Polyesters contain polar C=O and C-O ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar C-C and C-H bonds resistant to chemical and enzymatic attack (1).	[5]
Q48	Q48: 48. A* Challenge: Condensation Polymers & Biodegradability 48 48. (a) $\text{-O-CO-C}_6\text{H}_4\text{-CO-O-CH}_2\text{-CH}_2\text{-}$ with open continuation bonds at ends (2). 48. (b) Polyesters contain polar C=O and C-O ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar C-C and C-H bonds resistant to chemical and enzymatic attack (1).	[5]
Q49	Q49: 49. A* Challenge: Condensation Polymers & Biodegradability 49 49. (a) $\text{-O-CO-C}_6\text{H}_4\text{-CO-O-CH}_2\text{-CH}_2\text{-}$ with open continuation bonds at ends (2). 49. (b) Polyesters contain polar C=O and C-O ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar C-C and C-H bonds resistant to chemical and enzymatic attack (1).	[5]
Q50	Q50: 50. A* Challenge: Condensation Polymers & Biodegradability 50 50. (a) $\text{-O-CO-C}_6\text{H}_4\text{-CO-O-CH}_2\text{-CH}_2\text{-}$ with open continuation bonds at ends (2). 50. (b) Polyesters contain polar C=O and C-O ester bonds (1) susceptible to nucleophilic attack / hydrolysis by water and enzymes (1). Poly(ethene) contains strong non-polar C-C and C-H bonds resistant to chemical and enzymatic attack (1).	[5]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 15D**FAQ #1: Acyl Chloride vs Carboxylic Acid Esterification****Category: Ester Synthesis****COMMON EXAMINER TRAP / MISTAKE:**

Claiming carboxylic acid esterification is faster than acyl chloride.

MODEL ANSWER / EXAMINER EXPECTATION:

Acyl chlorides react VIGOROUSLY at room temperature with alcohols giving 100% yield (irreversible). Carboxylic acids require heating with conc H₂SO₄ catalyst and reach an equilibrium.

FAQ #2: Gaseous Product in Acyl Chloride Hydrolysis**Category: Observations****COMMON EXAMINER TRAP / MISTAKE:**

Stating chlorine gas is evolved when acyl chlorides react with water.

MODEL ANSWER / EXAMINER EXPECTATION:

Acyl chlorides evolve HYDROGEN CHLORIDE gas (HCl), which forms misty white fumes in moist air. Chlorine gas (Cl₂) is NOT formed.

FAQ #3: Saponification Irreversibility**Category: Alkaline Hydrolysis****COMMON EXAMINER TRAP / MISTAKE:**

Assuming alkaline hydrolysis of esters is reversible.

MODEL ANSWER / EXAMINER EXPECTATION:

Alkaline hydrolysis (saponification) forms a CARBOXYLATE ANION (RCOO⁻). The negative charge on RCOO⁻ prevents nucleophilic attack by the alcohol, driving the reaction 100% to completion.

FAQ #4: Polyester Repeat Unit Continuation Bonds**Category: Polymer Drawing****COMMON EXAMINER TRAP / MISTAKE:**

Omitting open bonds at the ends of polymer repeat units.

MODEL ANSWER / EXAMINER EXPECTATION:

Always extend open continuation bonds OUTSIDE the brackets of the repeat unit to show it is a repeating polymer chain.

FAQ #5: Biodegradability Mechanism**Category: Environmental Chemistry****COMMON EXAMINER TRAP / MISTAKE:**

Stating polyesters break down because they are soft.

MODEL ANSWER / EXAMINER EXPECTATION:

Polyesters are biodegradable because the C=O ester links are POLAR and can be HYDROLYSED by water and microbial enzymes. Addition polymers have non-polar C-C backbones resistant to hydrolysis.

FAQ #6: Acyl Chloride Reaction with Amines**Category: Amide Formation****COMMON EXAMINER TRAP / MISTAKE:**

Writing 1 mole of amine per mole of acyl chloride.

MODEL ANSWER / EXAMINER EXPECTATION:

Acyl chloride + 2 moles of primary amine $\rightarrow$ N-substituted amide + alkylammonium chloride salt. 2 moles of amine are needed because 1 mole reacts with HCl formed.

FAQ #7: Acid vs Alkaline Hydrolysis Products**Category: Hydrolysis Products****COMMON EXAMINER TRAP / MISTAKE:**

Writing carboxylic acid as the product of alkaline hydrolysis.

MODEL ANSWER / EXAMINER EXPECTATION:

Acid hydrolysis yields the CARBOXYLIC ACID (RCOOH). Alkaline hydrolysis yields the SODIUM CARBOXYLATE SALT (RCOONa). To get the free acid from saponification, acidify with dilute HCl.

FAQ #8: Acyl Chloride Preparation**Category: Reagents****COMMON EXAMINER TRAP / MISTAKE:**

Using dilute HCl to prepare acyl chlorides from carboxylic acids.

MODEL ANSWER / EXAMINER EXPECTATION:

Acyl chlorides are prepared using phosphorus(V) chloride PCl_5 , phosphorus(III) chloride PCl_3 , or thionyl chloride SOCl_2 with dry carboxylic acids.

FAQ #9: Chlorobenzene Unreactivity**Category: Aromatic Chlorides****COMMON EXAMINER TRAP / MISTAKE:**

Expecting chlorobenzene to react with water like acyl chlorides.

MODEL ANSWER / EXAMINER EXPECTATION:

Chlorobenzene $\text{C}_6\text{H}_5\text{Cl}$ is UNREACTIVITY towards nucleophilic substitution because p-orbitals on Cl overlap with the pi-cloud of the benzene ring, strengthening the C-Cl bond and increasing electron density.

FAQ #10: Transesterification Reactions**Category: Biodiesel Synthesis****COMMON EXAMINER TRAP / MISTAKE:**

Confusing transesterification with ester hydrolysis.

MODEL ANSWER / EXAMINER EXPECTATION:

Transesterification converts one ester into another by reacting it with an alcohol in the presence of an acid/base catalyst (e.g. vegetable oil + methanol $\rightarrow$ biodiesel + glycerol).